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Mean Absolute Directional Loss as a new loss function for machine learning problems in algorithmic investment strategies☆

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ABSTRACT

This paper investigates the issue of an adequate loss function in the optimization of machine learning models used in the forecasting of financial time series for the purpose of algorithmic investment strategies (AIS) construction. We propose the Mean Absolute Directional Loss (MADL) function, solving important problems of classical forecast error functions in extracting information from forecasts to create efficient buy/sell signals in algorithmic investment strategies. MADL places appropriate emphasis not only on the quality of the point forecast but also on its impact on the rate of achievement by the investment system based on it. The introduction and detailed description of the theoretical properties of this new MADL loss function are our main contributions to the literature. In the empirical part of the study, based on the data from two different asset classes (cryptocurrencies: Bitcoin and commodities: Crude Oil), we show that our new loss function enables us to select better hyperparameters for the LSTM model and obtain more efficient investment strategies, with regard to risk-adjusted return metrics on the out-of-sample data.

1. Introduction

The main idea for this paper comes from the unsolved dilemma regarding the search for forecasting models that can be used for buy/sell signal generation in algorithmic investment strategies (AIS). No matter what kind of theoretical concept is incorporated into the heart of such an investment model, we have a few similar issues that have to be properly addressed to increase the probability of generating efficient signals on out-of-sample (OOS) data. Among many others, these include the architecture of testing various models (machine learning, econometric, macroeconomic, or statistical approaches), the structure of the walk-forward procedure (usually consisting of numerous training, validation, and testing periods of different lengths), hyperparameters tuning and parameters optimization, model estimation phase, and finally the appropriate set of time series with possibly diverse characteristics of their distributions. The point is that all of these problems have to be designed optimally to avoid potential over-fitting issues and find the best possible variant of the investment model.

The majority of papers undertaking the topic of testing AIS do not pay proper attention to these problems and focus only on the empirical testing of one or several selected investment models, on a single instrument, over quite short data periods, usually without explaining the details of the whole procedure. In this paper, we take into account all these drawbacks and compare them. Nevertheless, the main focus is be put on one crucial aspect of testing such investment models, which is, in our opinion, the selection of a proper loss function. In reality, it has the greatest impact on hyper-parameter tuning, followed by the model estimation phase, and therefore it heavily affects the final results of AIS tests within such an architecture. Consequently, the introduction of the MADL loss function and the description of theoretical properties and its differences in comparison to classical loss functions (MSE, MAE, etc.) can be regarded as the main important contribution and novelty of this study.

The main hypothesis verified in this paper is as follows (RH): *The MADL loss function has better properties than classical forecast error*

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functions in the optimization of ML models used in forecasting financial time series for the purpose of AIS construction. When talking about the properties of our MADL function, we mean that by appropriately approaching the relationship: *the sign of the forecast in relation to the sign of the realized rate of return versus the absolute value of the realized rate of return*, we can create more effective investment systems in the context of their results in the out-of-sample period

In order to verify this hypothesis, after a brief reference to the most common drawbacks in papers testing AIS (Literature Review section), and a description of our architecture of testing (Methodology and Data section), we define and describe the new formula of the MADL function, focusing primarily on its conceptual differences to other similar functions. Then, in the Results section, we show the empirical comparison of the use of the MADL function and the classical forecast error function (MAE) on two time series: Bitcoin and Crude Oil daily simple returns. In this analysis, we visualize potential differences in a more realistic way than only theoretical deliberations. Finally, we conclude and present some extensions of this research.

2. Literature review

The literature review presented below focuses on critical drawbacks in papers that describe the testing of algorithmic investment strategies. The main issue is that the vast majority of these papers do not maintain a proper testing structure, which is why their results cannot be treated as valid and robust, even though the literature on this topic is very broad. Before moving on to the main part of this study, it is crucial to list and describe the most frequent flaws in papers examining different AIS. These include:

- Over-optimization of machine learning models ([1,2].
- Wrong optimization criteria or loss functions, including RMSE, MSE, MAE, MAPE, percent of over-predictions (%OP), and others. These functions are used by authors in a vast majority of publications, making it hard to choose the optimal methods for producing buy/sell signals [3,4].
- Only one in-sample period and one out-of-sample period, making the results highly dependent on the chosen period. A great number of research papers on this topic [1,2,5,6] highlight this issue as being very prevalent.
- No out-of-sample period, which results in a lack of forecasting ability of the paper's findings [7,8].
- As there is only one basis instrument used for AIS testing, the distributional characteristics of this instrument have a strict influence on the results [9].
- Forward-looking bias in buy/sell signals, which is typically caused by the use of future macroeconomic data or mistakes in the definitions of buy/sell signals [10–12]
- Lack of sensitivity analysis, which is essential in assessing the robustness of the model's results, with regard to the parameters that were initially chosen [3,13], and [4].
- Data snooping bias — when authors publish only their best results, without conducting a systematic examination of other parameters and assumptions [10,14].
- Survivorship bias — the selection of current index constituents, for instance, in research using data from the previous 20 years, is the most prevalent illustration of this bias [11].
- Inadequate performance metrics or their incorrect interpretation — the efficiency of investment strategies is evaluated only on the quality of point forecasts generated using the theoretical models which form the basis of such investment strategies, rather than of proper risk-adjusted return metrics calculated on the equity curves generated by these strategies [15,16].

Some of the above-mentioned drawbacks can be mitigated, at least in part, by pertinent testing of the model architecture. Special attention has to be paid to the hyperparameter tuning phase, where an appropriately selected loss function is critical.

3. Methodology and data

3.1. LSTM model

Long Short-Term Memory (LSTM) networks, depicted in Fig. 1, represent a variant of recurrent neural networks (RNNs) capable of preserving prolonged dependencies within data and mitigating the vanishing gradient issues commonly encountered in traditional RNNs. They find extensive application in modeling sequential data such as text, speech, and time series. LSTM units consist of memory cells, each equipped with three distinct gates: the input gate, output gate, and forget gate. These gates employ "tanh" and "sigmoid" functions to control the information flow within the cell, determining the amount and relevance of information to retain in the long-term state, transmit to the subsequent step, or discard.

The LSTM introduces a mechanism to maintain information (c_t) over multiple time steps, thereby preventing the gradual disappearance of older signals during processing. This information c_t is integrated with both the input and recurrent connections:

$$\text{output}_t = f(\text{state}_t \bullet U_o + \text{input}_t \bullet W_o + c_t \bullet V_o + b_o) \tag{1}$$

The new value of c_{t+1} is then calculated as:

$$c_{t+1} = i_t \cdot k_t + c_t \cdot f_t \tag{2}$$

where:

$$i_t = f(state_t \bullet U_i + input_t \bullet W_i + b_i) \tag{3}$$

$$f_t = f(state_t \bullet U_f + input_t \bullet W_f + b_f) \tag{4}$$

$$k_t = f(state_t \bullet U_k + input_t \bullet W_k + b_k) \tag{5}$$

and U_i, W_i, b_i, U_f, W_f, b_f, U_k, W_k, b_k are matrices with weights, $f(\cdot)$ is the activation function and $\bullet$ is the dot product of two vectors.

3.2. Mean absolute directional loss function

We introduce a novel loss function to address one of the most prevalent issues with papers testing algorithmic trading strategies. In this study, we use it to enhance the predictive power of the LSTM model [18] for these strategies.

Based on [19], we observe that popular classical error metrics like RMSE, MSE, MAE, MAPE, and %OP, used in the majority of similar studies, are inappropriate for assessing the effectiveness of the models' forecasting abilities in AIS. The reason for this is that the error metrics mentioned above do not consider the forecasting ability of investment signals that are based on these forecasts; rather, they only consider the forecasting accuracy of forecasts (i.e., the difference between the forecasted and observed value). It implies that almost all of these error metrics (RMSE, MSE, MAE, and MAPE) penalize forecast errors regardless of whether they are positive or negative (forecast error = $\hat{R}_i - R_i$), and their sign (sign($\hat{R}_i$) versus sign(R_i)), whereas the %OP metric only considers the forecast error's direction and not its magnitude. For this reason, researchers in the majority of other papers choose the signal combination (and train and validate their models accordingly) that optimizes only the chosen error metric and finds the models with the best point forecast rather than the models with the most profitable combination of signals for tested investment strategy.

To solve this problem, we suggest a new loss function, called Mean Absolute Directional Loss (MADL), which can be computed as:

$$\text{MADL} = \frac{1}{N}\sum_{i=1}^{N}(-1) \times \text{sign}(R_i \times \hat{R}_i) \times \text{abs}(R_i), \tag{6}$$

where MADL is the Mean Absolute Directional Loss, R_i is the observed return on interval i, $\hat{R}_i$ is the predicted return on the interval i, sign(X) is the function that returns -1,0,1 as the sign of X, abs(X) is the

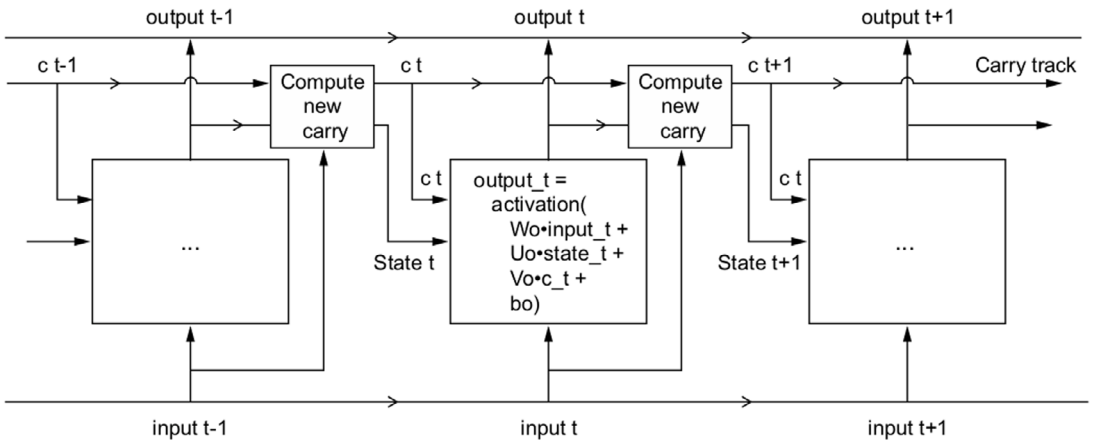


Fig. 1. Anatomy of the LSTM model. Note: LSTM cells presented in this figure show the information flow between the main LSTM gates: input, output, and forget. *Source:* [17].

function that gives the absolute value of X and N is the number of forecasts. In this manner, the value the function returns is equal to the observed return on investment with the predicted direction, allowing the model to determine whether the prediction results in profit or loss and the observed (real) amount of this profit or loss.

It is important to note here, that the absence of the value of $\hat{R}_i$ in Eq. (6) is intentional. The value of $\hat{R}_i$ is not needed in the properly constructed loss function because in reality, while designing algorithmic investment strategies, the only thing that matters for us is the proper direction of our prediction (sign($R_i \times \hat{R}_i$)), especially on days when the abs(R_i) was at relatively high levels.

MADL is designed specifically for generating signals for AIS. In our model, this function is minimized, ensuring that the strategy produces a profit if it returns negative values and a loss if it returns positive values. MADL is also the main loss function used in hyperparameters tuning and in the estimation of the LSTM model.

3.3. The main differences between MADL and MAE

In order to reveal the properties of the MADL function and visualize its differences with regard to classical error metrics, (e.g. MAE explained in Eq. (7)) we present its main distinguishing characteristics, shown in three consecutive figures (Fig. 2, 3, and 4).

$$\text{MAE} = \frac{1}{N}\sum_{i=1}^{N} \text{abs}(R_i - \hat{R}_i), \tag{7}$$

where R_i is the observed return on interval i, $\hat{R}_i$ is the predicted return on the interval i.

The reason why we chose MAE for the comparison and in the empirical part of the study is because its concept is closest to the proposed MADL function. Both MADL and MAE utilize absolute values, and their simplicity allows us to illustrate the differences between the functions in the clearest manner. However, the comparisons presented below can also be expanded to other types of functions (squared, root squared, and percentage).

Fig. 2 depicts the difference between MAE and MADL logic that is responsible for the large differences between these two loss functions.

Fig. 3 presents the difference between MAE and MADL logic in the case when the difference is relatively small but still affects the final values of the loss functions.

Fig. 4 presents the difference between MAE and MADL logic in the case when predicted and observed returns have an opposite sign and their values are significantly different.

Fig. 2, 3, and 4 show the most important conceptual differences between classical error functions and MADL. Additionally, Fig. 5 presents the distribution of error values in the case of large and small differences between observed return (R_i) and predicted return ($\hat{R}_i$) or zero. What is significant about the MADL loss function is that the reference point for the loss measurement is set to zero, whereas in almost all other classical error metrics, it is the predicted return ($\hat{R}_i$). Fig. 5 presents the comparison of MAE and MADL for various values of predicted and observed returns.

3.4. Data

We use simple returns based on daily data of two assets from various asset classes: Bitcoin (BTC) and Crude Oil (represented by ETF named USO). The data we used cover the period from July 7, 2010 (for BTC) and April 18, 2007 (USO), until March 31, 2022 (for both assets). Sources of the BTC data were Kraken, Bitfinex, BTC-e, CEX, and Coinbase exchanges, while USO data was obtained from stooq.com. These returns serve as the input variable (single feature) and as an output target. The exact description of the division of data into sets and samples is presented in Table 1 and in the empirical procedure described in Section 3.5.

3.5. Research description

It is important to describe the proper architecture for training and testing ML models used for AIS construction. Although, the main aim of this paper is the presentation of the concept and main characteristics of the new loss function, the empirical part (see Section 4) with the testing example is added for visualization purposes and as an explanation of its use in practice. Fig. 6 presents the process used to explain how exactly our AIS was tested.

We use the LSTM model with a walk-forward procedure of training and testing, as it is considered one of the most successful and widely used deep learning models for time series forecasting. The hyperparameter tuning phase is performed during the initial in-sample period, followed by model training and estimation on the remaining data. The majority of the tuning is conducted manually. For this purpose, we use various combinations of the loss functions to extract information about the usefulness of the selected loss function in such a procedure. Table 1 summarizes the values of hyperparameters selected in the tuning phase for each loss function separately.

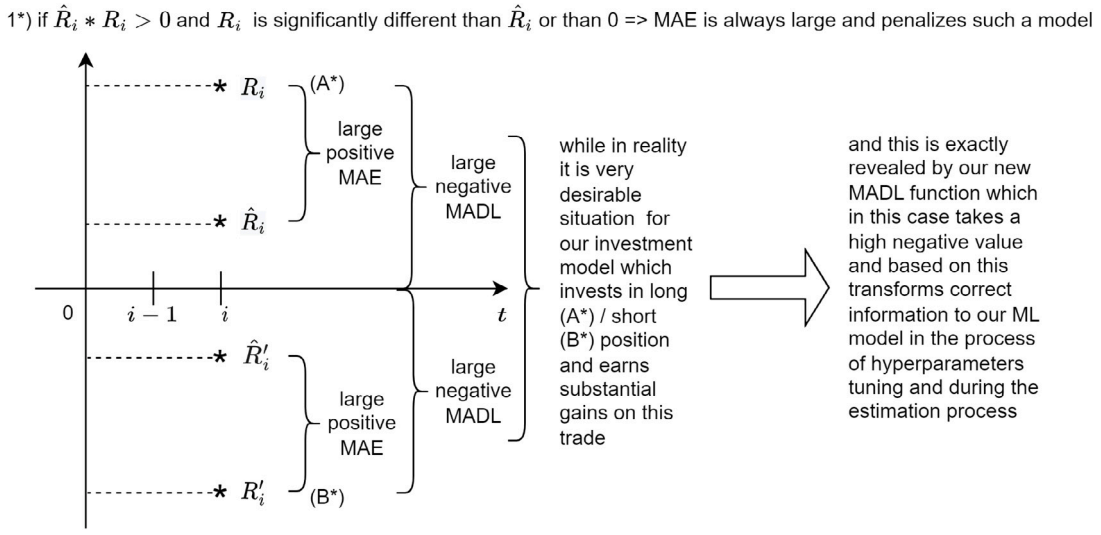


Fig. 2. The difference between MAE and MADL logic #1.
Note: Variants (A*), and (B*) show that in specific cases the difference between MAE and MADL values can be significant, and therefore affect the final selection of model parameters.

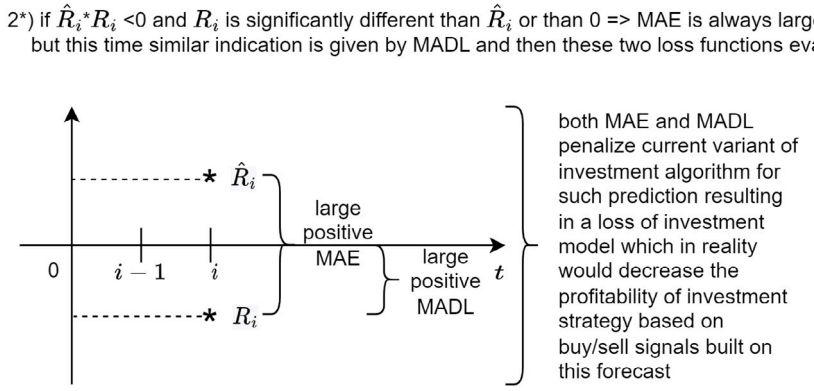


Fig. 3. The difference between MAE and MADL logic #2.
Note: This variant depicts the difference between MAE and MADL by treating the error as the distance between the predicted and observed values (MAE) and the observed and 0 (MADL) in cases where the sign of the prediction is not the same.

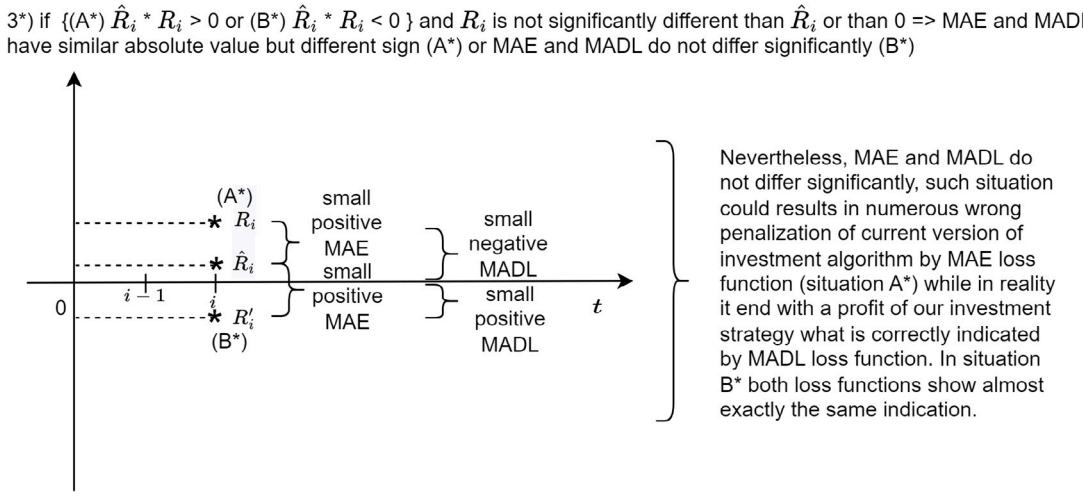


Fig. 4. The difference between MAE and MADL logic #3.
Note: The last example of the difference is negligible with regard to its value for a single prediction, but taking into account that we can have quite a few of them, it can substantially influence the aggregated value of the selected loss function.

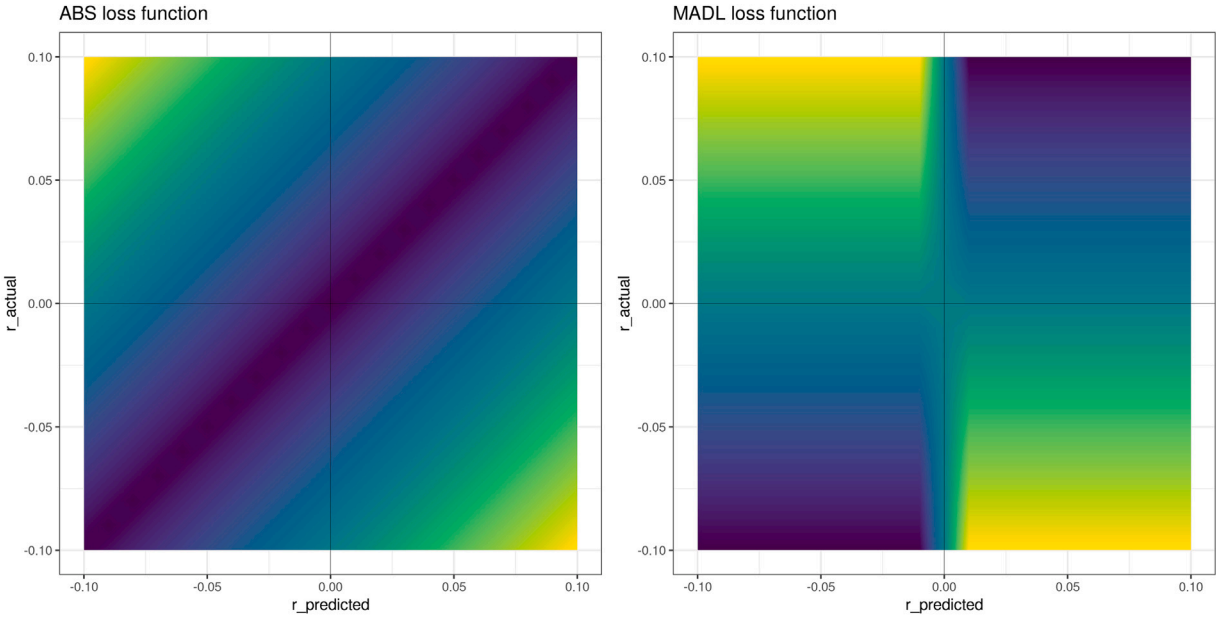


Fig. 5. Loss functions: MADL vs MAE.
Note: Dark blue color indicates the lowest error, while yellow color indicates the highest error. The left panel (with MAE) shows that the error is strictly dependent on the difference between $\hat{R}_i$ and R_i while the right panel shows that $\hat{R}_i$ is always compared with the zero reference point in the process of error calculation.

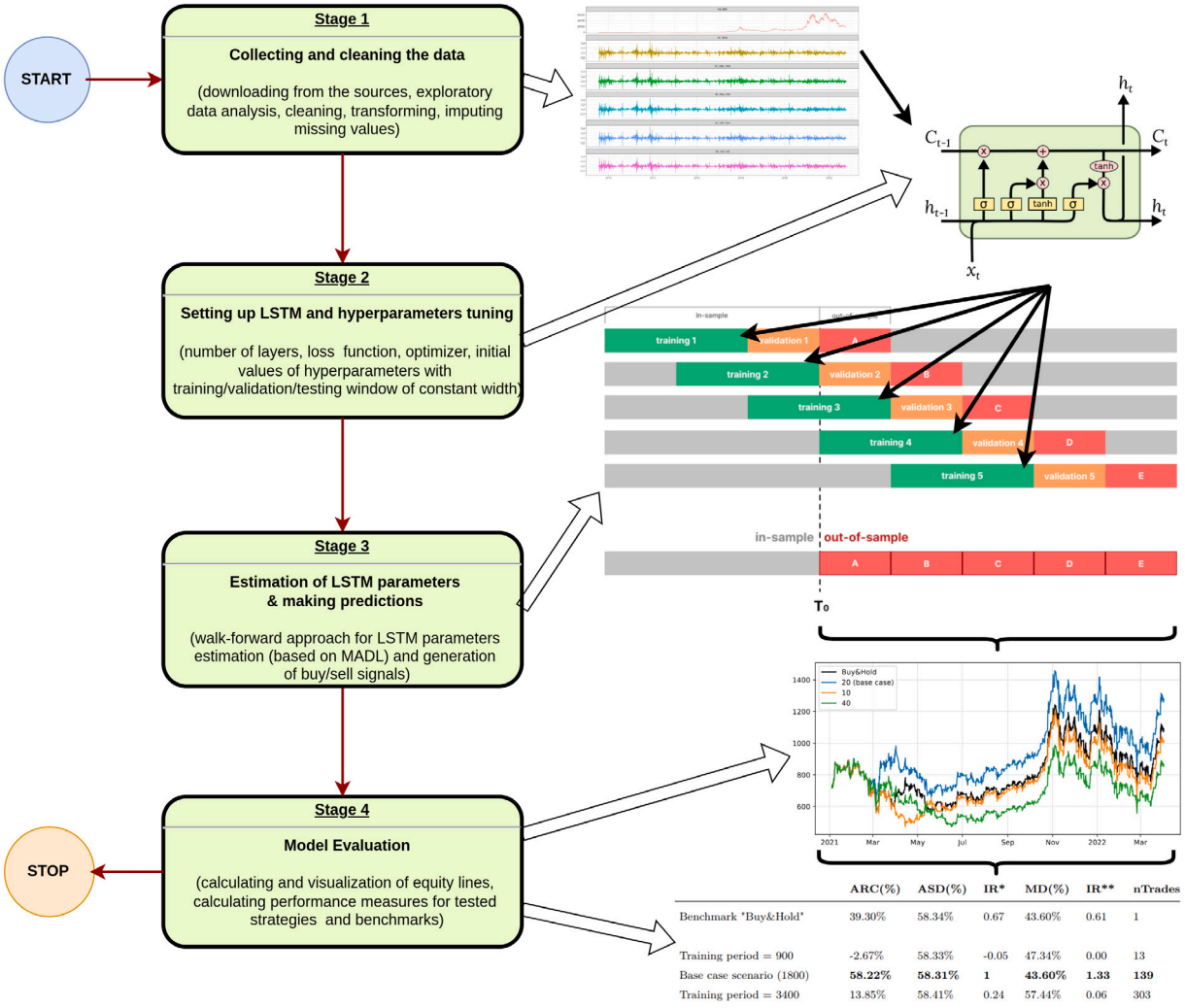


Fig. 6. The process of LSTM model testing with MADL function. Note: The MADL function is used in stages 2 and 3.

Table 1
Selected values of hyperparameters.

Hyperparameter	Selected value MADL	Selected value MAE
No. hidden layers	3	3
No. neurons	512/256/128	64/32/16
Activation function	tanh	tanh
Recurrent Activation	sigmoid	sigmoid
Dropout rate	0.02	0.0002
l2 regularizer	0.0005	0.00001
Optimizer	Adam	Adam
Learning rate	2.15	0.0015
BTC train/test	365-1460*/365	1460/365
USO train/test	252-1008*/252	1008/252
Batch size (BTC/USO)	1460/1008	1460/1008
Sequence length (BTC/USO)	20/10	20/10
Epochs	300	200

Note: MADL hyperparameters (second column) are used for MADL/MADL and MADL/MAE strategies, while MAE hyperparameters are used for MAE/MAE and MAE/MADL strategies. * train set starts with one year and extends to four years.

Finally, we perform the following steps to conduct the empirical part of this research:

- The division of data into in-sample (training and validation) and out-of-sample (test) sets, each with 1460/365 observations for BTC and 1008/252 observations for Crude Oil. Train set extends from one year to up to four years, to assure the maximum possible out-of-sample periods.
- The size of the validation set is 33% of the train set.
- Hyperparameters tuning based on the proposed MADL function, described in Section 3.2 and the classic MAE loss function.
- The Walk-forward (rolling window) approach is used for training and testing. During a single iteration, the model is trained and validated using four years of data. The optimal weights of the model are then used to create predictions on the test set (equal to one year of data). The window is then moved by one year forward, the weights are reset and the model is retrained (see Fig. 6, Step 3).
- Each prediction is made using 20 (for BTC) and 10 (for USO) past observations, which is equal to the sequence length hyperparameter (see. Table 1.
- Buy/Sell signals are generated based on the sign of next-day forecasts.

$$\begin{cases} \text{if } (\text{MADL}_{t+1}) > 0 & => \quad \text{BUY}_{signal} \\ \text{if } (\text{MADL}_{t+1}) < 0 & => \quad \text{SELL}_{signal} \end{cases} \tag{8}$$

- Tests for one type of strategy: Long/Short.
- Equity curves and performance metrics for tested AIS from the two loss functions and assets according to [20]. These results are provided in Section 4.

In the empirical part of the research, our focus lies in demonstrating the model's performance using MADL, and we further compare it with MAE to maintain consistency with the theoretical introduction (refer to Section 3.3). While the comparison with other types of loss functions (such as MSE, MAPE, or classification functions) is not the primary objective of this research, it could be pursued similarly. However, it is important to note that re-tuning of hyperparameters would be necessary for each new loss function, a factor that could potentially impact the results. To illustrate this impact we present the results (see Table 3) in combination with hyperparameters tuned using MADL (MADL/MADL and MADL/MAE) and tuned using MAE (MAE/MADL and MAE/MAE).

3.6. Performance metrics

In order to evaluate the efficiency of tested strategies, we calculate the following performance metrics based on [21,22]. The performance metrics are divided into four categories:

- Return performance metrics: Annualized Return Compounded (ARC)
- Risk performance metrics: Annualized Standard Deviation (ASD), Maximum Drawdown (MD), Maximum Loss Duration (MLD)
- Risk-adjusted return performance metrics: Information Ratio (IR*), Modified Information Ratio (IR**), Aggregated Information Ratio (IR***)
- Informative performance metrics: Number of observations (nObs), Number of trades (nTrades).

4. Results

Table 2 presents the results of the LSTM model used in this research, for which the hyperparameters tuning, training, and estimation are performed based on various combinations of loss functions: MADL/MADL, MADL/MAE, MAE/MAE, and MAE/MADL.[1] The performance metrics for final equity curves show that no matter which risk-adjusted return metrics (IR*, IR**, and IR***) we select, the result is always the best in the case of each asset for the MADL/MADL approach.

Fig. 7 presents the fluctuations of equity curves for BTC and USO in two panels (top: BTC and bottom: USO). Fig. 7 which shows the best results for MADL/MADL can be treated as an additional confirmation of numeric results presented in Table 2.

Table 3 presents an attempt to test the statistical significance of the results focusing on α and β coefficients from the regression in the form of $R_t = \alpha + \beta R_t^* + \varepsilon_t$, where R_t is the buy and hold returns, and R_t^* returns from Long/Short strategy. The results presented in Table 3 confirm those presented in Table 2, with positive and significant α for MADL/MADL strategy in case of BTC and confirm only partly MADL/MADL strategy for USO (positive α but not significant).

5. Conclusions

This paper aims to introduce the new concept of a more accurate loss function, which is appropriately adjusted to the problem of the optimization of machine learning models used in the forecasting of financial time series, and for the construction of algorithmic investment strategies (AIS). For this purpose, we propose a new loss function — MADL (Mean Absolute Directional Loss), presented the rationale for its construction, and a detailed interpretation with special attention to its difference with regard to classical loss functions used in the literature. Last but not least, we demonstrate how the new loss function helps us choose better hyperparameters for our ML models and produce more effective investment strategies in terms of risk-adjusted return metrics on the out-of-sample data using two different asset classes (cryptocurrencies: Bitcoin and commodities: Crude Oil).

Future research could focus on modifying the original MADL loss function by introducing different penalization structures, such as "squared" instead of "absolute," which might help with the challenges in optimizing the function's gradient. This modification has the potential to improve the training of various machine learning models responsible for generating buy/sell signals in the forecasting process.

[1] MADL/MADL indicates that the hyperparameters are tuned using MADL, and subsequently, the model is trained and tested using MADL. MADL/MAE denotes that the same hyperparameters are employed to evaluate the model with MAE. Similarly, in the instances of MAE/MAE and MAE/MADL, the methodology remains consistent.

Table 2
Comparison of two approaches.

Model	aRC	aSD	MD	MLD	IR*	IR**	IR***	nObs	nTrades
				Panel A — BTC					
B&H	91.25	87.37	86.67	3.24	1.04	1.100	0.310	4107	2
MADL/MADL	109.94	87.34	75.04	1.91	1.26	1.844	1.062	4107	306
MADL/MAE	99.90	87.36	86.67	3.24	1.14	1.318	0.406	4107	4
MAE/MAE	44.26	87.45	94.18	3.46	0.51	0.238	0.030	4107	330
MAE/MADL	4.05	87.51	90.99	4.83	0.05	0.002	0.000	4107	320
				Panel B — USO					
B&H	−10.59	38.09	98.19	14.24	−0.28	−0.030	0.000	3901	2
MADL/MADL	5.25	38.08	62.73	7.58	0.14	0.012	0.000	3901	731
MADL/MAE	−22.70	38.07	98.91	15.42	−0.60	−0.137	−0.002	3901	11
MAE/MAE	−17.02	38.08	96.88	15.42	−0.45	−0.079	−0.001	3901	402
MAE/MADL	4.19	38.08	84.39	12.74	0.11	0.005	0.000	3901	692

Note: Panel A presents results for BTC, while panel B presents the results for USO.

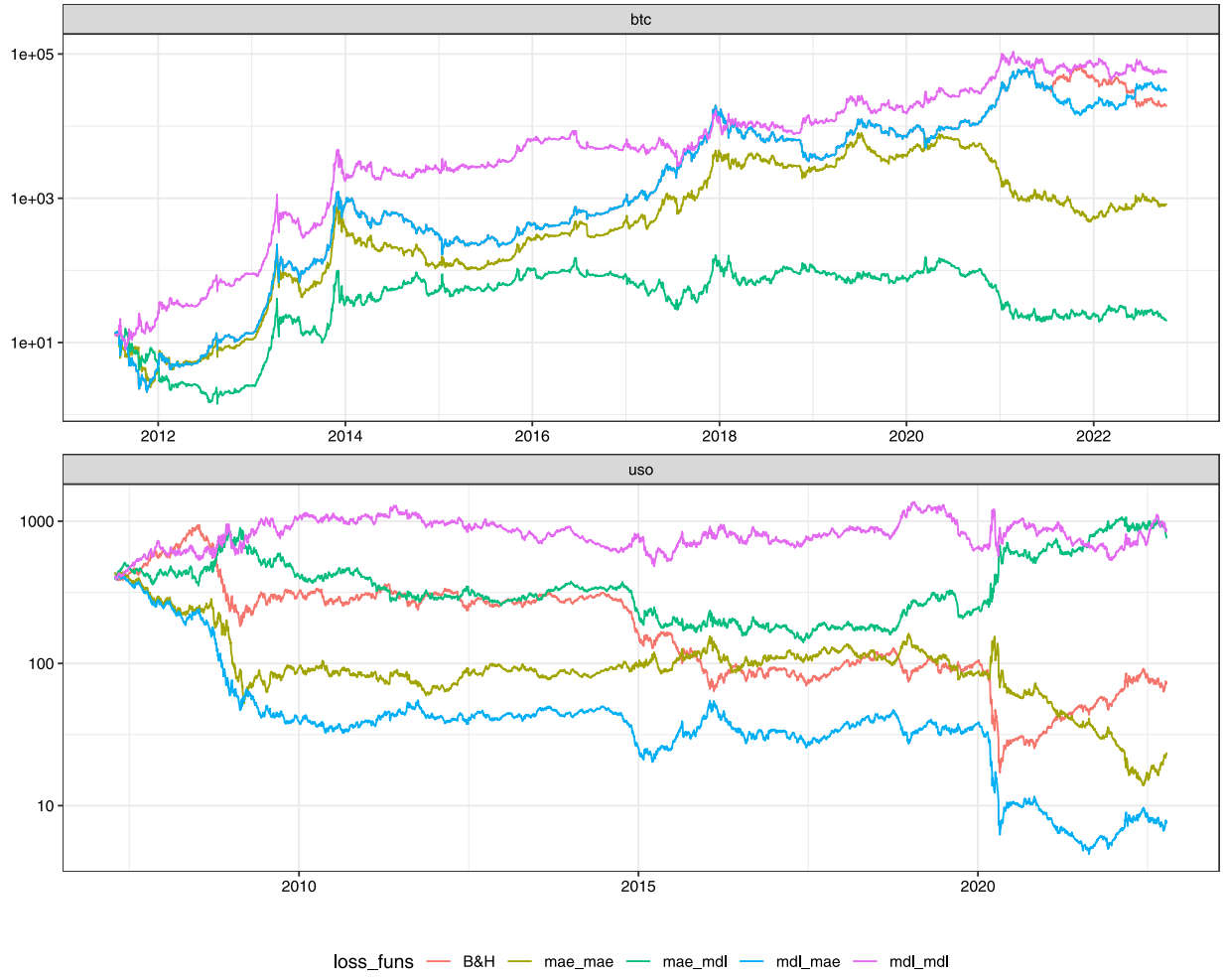


Fig. 7. Equity curves.
Note: Equity curves present the fluctuations of investment strategies for BTC (upper panel) and USO (lower panel) for strategies trained based on various combinations of loss functions used (MAE/MAE, MAE/MADL, MADL/MAE, MADL/MADL) in the period between July 7, 2011, and December 31, 2021 (for BTC) and in the period between April 18, 2008, and December 31, 2021 (for USO). Additionally, we included the Buy & Hold (B&H) strategy as a benchmark strategy for comparison purposes.

Table 3
Results of regressions for returns: B&H vs. MAE/MAE, MAE/MADL, MADL/MAE, MADL/MADL.

Dep. var.	Alpha	StdErr	t	pv	Beta	StdErr	t	pv
				Panel A — BTC				
MAE/MAE	0.0009	0.001	1.431	0.152	0.394	0.014	27.4	0.000
MAE/MADL	0.0007	0.001	1.014	0.310	0.159	0.015	10.2	0.000
MADL/MAE	0.0004	0.000	1.307	0.191	0.895	0.007	129.0	0.000
MADL/MADL	0.0029	0.001	4.027	0.000	0.073	0.016	4.6	0.000
				Panel B — USO				
MAE/MAE	−0.0005	0.000	−1.423	0.154	−0.374	0.015	−25.1	0.000
MAE/MADL	0.0004	0.000	1.104	0.269	−0.236	0.016	−15.1	0.000
MADL/MAE	−0.0006	0.000	−1.956	0.050	0.491	0.014	35.1	0.000
MADL/MADL	0.0004	0.000	1.215	0.224	−0.228	0.016	−14.6	0.000

Note: The table presents the results of regressions in the form of $R_t = \alpha + \beta R_t^* + \varepsilon_t$, where R_t is the return for tested strategy in period t and R_t^* is the return in of BTC or USO strategies. Regressions were calculated between July 7, 2011, to December 31, 2021 (for BTC) and between April 18, 2008, and December 31, 2021 (for USO). The hyperparameters of the LSTM model for the base case scenario were set as was described in Table 1.

CRediT authorship contribution statement

Jakub Michańków: Writing – review & editing, Writing – original draft, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Paweł Sakowski:** Writing – review & editing, Writing – original draft, Visualization, Validation, Project administration, Methodology, Data curation, Conceptualization. **Robert Ślepaczuk:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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